

Vincent Lin

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EDUCATION

University of Florida - Gainesville, FL

May 2027

Bachelor of Science in Computer Science, Minor in Electrical Engineering

GPA: 3.62

- Relevant Coursework: Operating Systems, Databases, Programming Language Concepts, Data Structures and Algorithms, Computer Organization, Application of Discrete Structures

TECHNICAL SKILLS

Languages: C++, Python, JavaScript/TypeScript, Java, SQL

C++ & Systems: STL, Multithreading, TCP/UDP Sockets, POSIX, CMake, GDB, Valgrind

Tools & Platforms: Git, Linux, Docker, Firebase, Cloudflare, Jira, Figma, Quartus Prime, Arduino, Raspberry Pi

PROJECTS

Peer-To-Peer File Sharing System | C++, TCP Sockets, Multithreading, POSIX

Personal Project

- Architected a **P2P file transfer system** in C++ handling **10+** peer connections and files up to **4GB** with a custom protocol over **TCP sockets** with **SHA-256 hash verification** achieving **99.9%** data integrity.
- Designed **thread-safe** architecture using **mutexes** to coordinate piece scheduling across **8+** simultaneous peers, with state tracking that reduced redundant transfers by **~40%** and improved throughput to **50+ MB/s**

WAD Filesystem Daemon | C++, FUSE, POSIX, Linux

Personal Project

- Built a **FUSE-based userspace filesystem** in C++ that mounts WAD archive files as native Linux directories, implementing **8 syscall handlers** (open, read, write, mkdir, mknod, getattr, readdir, release)
- Developed a **tree-based inode structure** with O(1) path lookups via hash maps, supporting **buffered writes**, **file handle tracking**, and **atomic descriptor table updates** for crash-safe metadata persistence

Robot Movement Tracker | C++, OpenCV, Arduino, Serial Communication

Personal Project

- Created a **real-time** human tracking system in C++ using **OpenCV's DNN** detector processing **30 FPS** video at **720p resolution**, achieving **85%** detection accuracy while tracking up to **5 simultaneous targets**
- Engineered a **servo control system** with coordinate transformation algorithms mapping pixel positions to pan/tilt angles, featuring **PID** smoothing achieving **<100ms latency** and **<5% overshoot**

CaseForward AI | Next.js, React, TypeScript, MongoDB, Cloudflare

1st Place SwampHacks XI

- Designed a **serverless multi-agent AI system** using Google **Gemini 2.5** orchestrating **3** specialized agents to parse and deliberate different mediums, generating attorney-reviewable recommendations with **90%** confidence
- Featured **zero-trust storage** on **Cloudflare R2+Workers** with **client-side encryption**, IP allowlisting, and **metadata-only database** architecture securing **10GB+** of sensitive documents

Sociopolis | React, TypeScript, Firebase, Agile

Personal Project

- Collaborated on a **4-person Agile team** to build a full-stack application, owning **Firebase backend** integration including **data schemas**, **access patterns**, and **API contracts** enabling parallel frontend development

EXPERIENCE

Software Engineering Intern | Tuttokool

May 2025 – Aug. 2025

- Developed purchasing and warranty features for an e-commerce platform processing **\$10,000+ daily transactions** from **1,000+ professional contractors**, implementing order management and product browsing workflows
- Deployed a virtualized NVR on **Proxmox** supporting **150+ IP cameras** across **12** retail locations with **RAIDZ2** storage protecting **100TB** of video archives, recovering **\$15,000** in collision-related insurance claims

LEADERSHIP

Historian | Society of Asian Scientists & Engineers (SASE)

June 2025 – Present

- Grew UF SASE's Instagram (@ufsase) to **~3,400 followers** by documenting **80+ chapter events** and producing content including an annual promotional video with **10,000 views** and reels averaging **5,000 impressions**
- Mentored a **10-person** committee to plan and execute a general body meeting for **200 attendees** while contributing to **30+ resume reviews** and coordinating a **40+ person headshot** fundraiser raising **\$200+**